

Shipment Risk Analysis & Late Delivery Prediction

➤ Project Report

1. Introduction

In modern supply chain systems, timely delivery plays a critical role in maintaining customer satisfaction and operational efficiency. However, delays in shipments can lead to increased costs, reduced customer trust, and inefficiencies in logistics planning.

This project focuses on analyzing shipment data to identify patterns behind delivery delays and building a predictive model to forecast late deliveries. By combining machine learning with business intelligence, the project aims to shift from reactive problem-solving to proactive decision-making.

2. Problem Statement

A significant portion of shipments are delivered late, but businesses often lack a system to predict these delays in advance. Without predictive insights, organizations cannot take preventive actions.

The key problem addressed in this project is:

How can we identify high-risk shipments before dispatch and reduce late deliveries using data-driven insights?

3. Objectives

- Analyze shipment data to identify delay patterns
 - Build a machine learning model to predict late deliveries
 - Categorize shipments into risk levels (High, Medium, Low)
 - Develop an interactive Power BI dashboard
 - Provide actionable business recommendations
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4. Dataset Overview

The dataset used is the DataCo Smart Supply Chain dataset.

- Total Records: 180,519
- Features: 50+

- Key attributes include:
 - Shipping Mode
 - Delivery Status
 - Order and Shipping Dates
 - Product Category
 - Sales and Profit
 - Customer and Regional Information

The dataset provides a comprehensive view of supply chain operations.

5. Data Preprocessing

Key steps performed:

- Removed irrelevant columns (e.g., Product Description)
 - Handled missing values:
 - Customer Zipcode filled with median
 - Order Zipcode dropped due to high missing values
 - Converted date columns to datetime format
 - Created new feature:
 - **Shipping Delay = Actual Days - Scheduled Days**
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6. Exploratory Data Analysis (EDA)

Key Findings:

- Late Delivery Rate: **38.29%**
- High Risk Shipments: **10,110**
- Majority shipments (61.7%) fall under Low Risk

Insights:

- Standard Class contributes the highest delay volume
- Same Day shipping has the best performance
- Certain product categories show consistently high delays
- Regional differences significantly impact delivery performance

7. Feature Engineering

Features used for modeling:

- Days for shipment (scheduled)
- Sales per customer
- Benefit per order
- Order item quantity
- Shipping Mode (encoded)
- Order Region (encoded)
- Category Name (encoded)

Target Variable:

- Late Delivery Flag (1 = Late, 0 = Not Late)

8. Machine Learning Model

Models Used:

1. Logistic Regression (baseline model)
2. Random Forest Classifier (final model)

Why Random Forest?

- Handles non-linear relationships
- Works well with mixed data types
- Provides feature importance
- More robust than simple models

9. Model Performance

Accuracy: ~70%

Confusion Matrix Summary:

- Good at identifying non-late deliveries
- Moderate performance in detecting late deliveries

Key Insight:

The model is reliable for identifying high-risk shipments but still has room for improvement in recall.

10. Feature Importance

Top contributing features:

- Days for shipment (scheduled)
- Shipping Mode
- Benefit per order
- Sales per customer
- Order quantity

Interpretation:

Scheduling and shipping mode are the most critical factors influencing delays.

11. Risk Categorization

Predicted probabilities were converted into business-friendly categories:

- High Risk → Probability > 70%
- Medium Risk → 40% – 70%
- Low Risk → < 40%

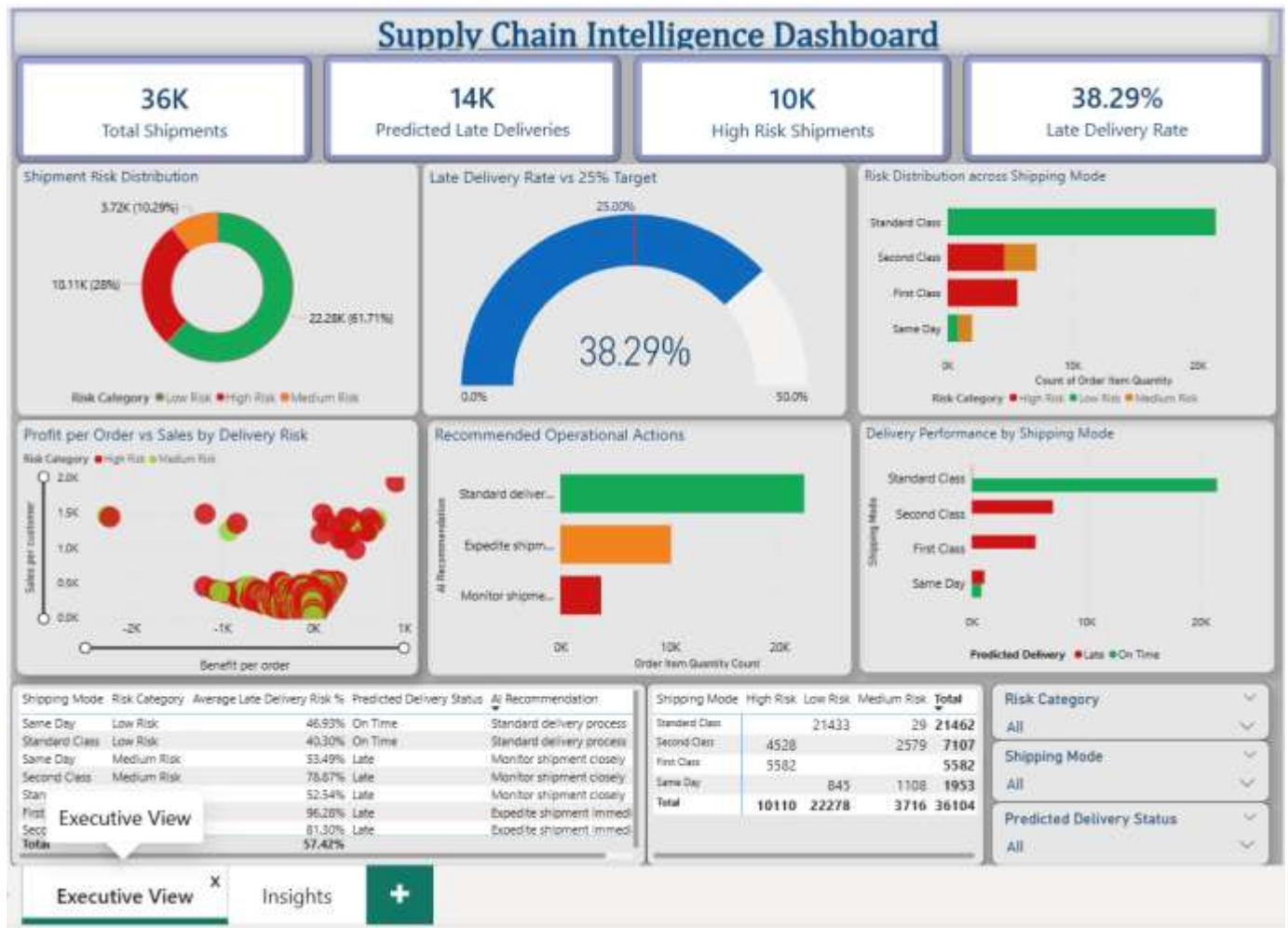
Distribution:

- High Risk: 10,110
 - Medium Risk: 3,716
 - Low Risk: 22,278
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12. Power BI Dashboard

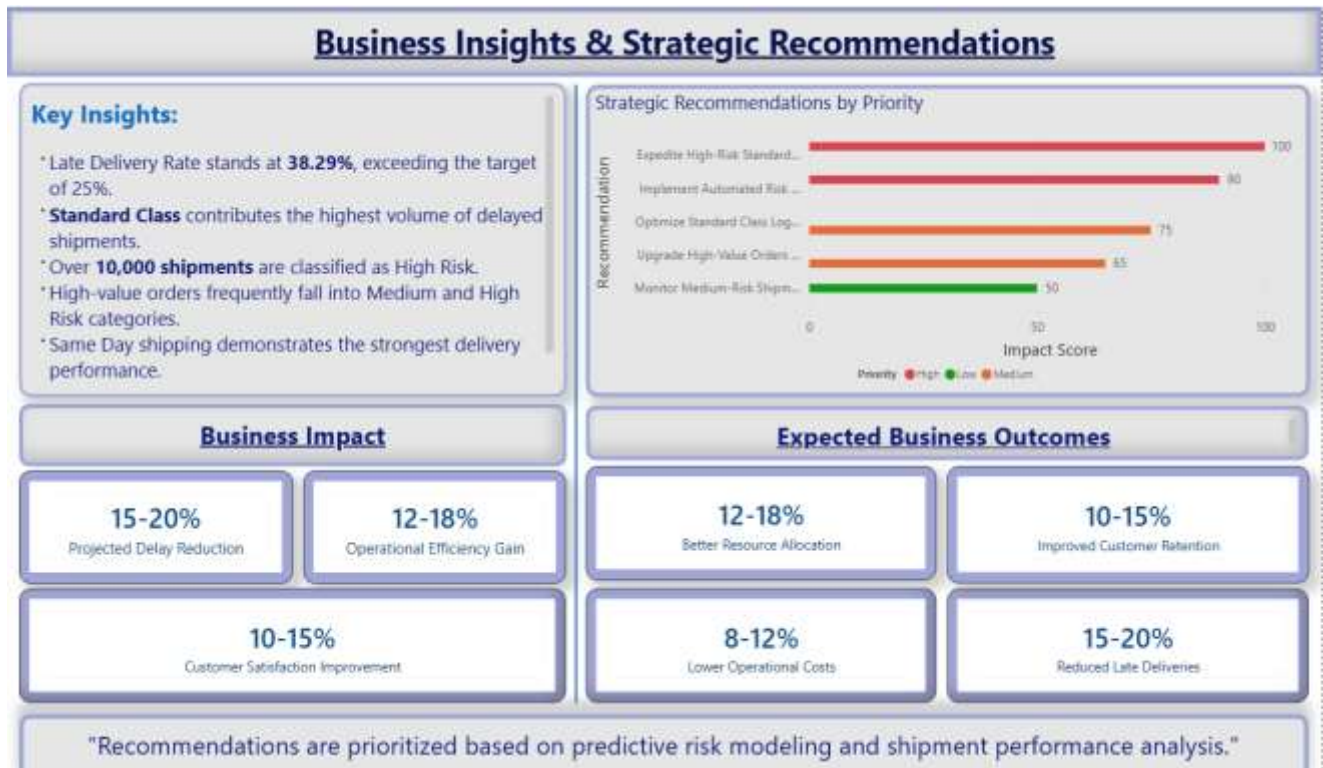
Page 1: Executive Overview

- Shipment KPIs
- Late Delivery Rate Gauge
- Risk Distribution
- Shipping Mode Analysis
- Regional Performance



Page 2: Insights & Strategy

- Key Insights
- Strategic Recommendations
- Business Impact
- Expected Outcomes



13. Key Insights

- Late delivery rate exceeds the target benchmark
- Standard Class shipments are the primary source of delays
- Over 28% of shipments are high risk
- Medium risk shipments present opportunity for improvement
- Same Day shipping shows strong reliability

14. Business Recommendations

- Prioritize high-risk shipments
- Optimize Standard Class logistics
- Implement automated risk alerts
- Upgrade shipping mode for high-value orders
- Monitor medium-risk shipments closely

15. Expected Business Impact

- Delay Reduction: 15–20%
 - Operational Efficiency Gain: 12–18%
 - Cost Reduction: 8–12%
 - Customer Satisfaction Improvement: 10–15%
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16. Conclusion

This project demonstrates how data analytics and machine learning can be applied to solve real-world supply chain problems. By predicting shipment delays and identifying risk factors, businesses can take proactive measures to improve efficiency and customer satisfaction.

The integration of predictive modeling with an interactive dashboard ensures that insights are not only generated but also effectively communicated for decision-making.

17. Future Scope

- Real-time shipment monitoring
 - Deployment using Streamlit or Flask
 - Integration with logistics systems
 - Advanced models like XGBoost
 - Automated alert systems
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18. Tools & Technologies

- Python
 - Pandas, NumPy
 - Scikit-learn
 - Matplotlib, Seaborn
 - Power BI
 - Jupyter Notebook
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20. Acknowledgment

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